Sec: SR.IIT_*COSC(MODEL-A) | GTM-7 | Date: 03-01-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	D	2	D	3	D	4	B	5	C
6	D	7	B	8	A	9	A	10	C
11	C	12	D	13	A	14	B	15	B
16	B	17	C	18	B	19	A	20	A
21	750	22	2600	23	517	24	875	25	4

CHEMISTRY

26	B	27	D	28	A	29	B	30	C
31	A	32	A	33	C	34	C	35	D
36	D	37	D	38	A	39	C	40	C
41	B	42	A	43	B	44	D	45	A
46	23	47	2	48	900	49	7	50	4

MATHEMATICS

51	D	52	B	53	D	54	B	55	D
56	D	57	B	58	B	59	C	60	B
61	D	62	A	63	C	64	B	65	B
66	A	67	C	68	A	69	C	70	D
71	5	72	4	73	2	74	60	75	31

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SOLUTIONS

PHYSICS

6. $C' = \frac{C}{2} \Rightarrow Q' = \frac{C}{2}V$

But $Q = CV \Rightarrow \Delta Q = -\frac{C}{2}V$

$\Rightarrow W = -\frac{1}{2}CV^2$

19. $B = \mu_0 H = 4\pi \times 10^{-7} \times \frac{2}{\pi} \times 10^7 = 8T$

$\delta = \frac{V^2}{aN} = \frac{V^2 m}{qVR\sin\theta} = \frac{mV}{qB\sin\theta}$

$= \frac{1 \times \sqrt{3} \times 2}{1 \times 8 \times \sqrt{3}} = \frac{1}{4}m$

CHEMISTRY

36. Platinum (II) not platinate (II) as it is neutral complex.

39. $\omega_{12} \rightarrow$ Isothermal Expansion

$\omega = -nRT\ln\frac{v_2}{v_1}$

$= -1 \times 2 \times 300 \times \ln\frac{20}{10}$

$= -1 \times 2 \times 300 \times \ln 2 \quad (\ln 2 = 0.7)$

$= -420\text{cal}$

$\omega_{23} \rightarrow$ Isobaric $(P = \text{ constant})$

$\omega = -P_{ext}\Delta V \quad (PV = nRT)$

$= -nR\Delta T$

$= -1 \times 2 \times 300 \quad (600K - 300K)$

$= -600\text{cal}$

$\omega_{34} \rightarrow$ Isothermal

$\omega = -\text{ nRT } \ln\frac{v_2}{v_1}$

$= -1 \times 2 \times 600 \times \ln\frac{10}{40}$

$= -1200 \times \ln 1 - 4$

$= -1200 \times \ln 1 - (2)^2$

$= -1200 \times \ln 1 - 2 \times 2 \quad (\ln 2 = 0.4 \Rightarrow \ln = 1.4)$

$= -1680\text{cal.}$

Net work done $= \omega_{12} + \omega_{23} + \omega_{34}$

$= -420 - 600 + 1680$

$= +660\ cal$

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45. Chromium(III)salt $\xrightarrow{\Delta} Cr_2O_3$

Brorax $\xrightarrow{\Delta} B_2O_3 + NaBO_2$

$2Cr_2O_3 + 6\,B_2O_3 \rightarrow 4Cr(BO_2)_3$

47. $\Delta Tf = ik_f m \; i = \dfrac{0.26}{1.86\times 0.1} = 1.3978 = 1.40$

$\alpha = \dfrac{i-1}{n-1} \therefore n-1 = \dfrac{1.4-1}{0.20}, n = 2+1 = 3 \therefore x = 2$

$H_2\,A \rightarrow 2H^+ + A^{2-}$

MATHEMATICS

51. NCERT

52. Let P be $(9\cos\theta, 3\sin\theta)$

$\Rightarrow$ For normal at maximum distance from origin $9\tan\theta = 3\cot\theta$

$\Rightarrow \tan\theta = \pm\dfrac{1}{\sqrt{3}} \Rightarrow m = \pm\dfrac{1}{\sqrt{3}}$

53. $f(x) = (x-2)^{-5} + (x-2)^{-4} + 3(x-2)^{-3} + 1 + (x-2)^8 + (x-2)^{10}$

Say $x-2 = t(>0)$

$f(x) = t^{-5} + t^{-4} + 3t^{-3} + 1 + t^8 + t^{10}$. Now apply $AM \geq GM$

$\dfrac{t^{-5}+t^{-4}+t^{-3}+t^{-3}+t^{-3}+1+t^8+t^{10}}{8} \geq \left(t^{-5}\cdot t^{-4}\cdot t^{-3}\cdot t^{-3}\cdot t^{-3}\cdot 1\cdot t^8\cdot t^{10}\right)^{\frac{1}{8}}$

$\Rightarrow f(x) \geq 8 \forall x > 2$ min value of $f(x)$ is 8

54. Given family of circles is $x^2 + y^2 - 4y + 3 + \lambda x = 0$

$\Rightarrow$ Common points are $(0,1)$ and $(0,3)$

Clearly $S' \equiv 0$ will be the circle with common points as diameter

$\Rightarrow S' \equiv (x-0)(x-0) + (y-1)(y-3) = 0$

$\Rightarrow x^2 + y^2 - 4y + 3 = 0$

$\Rightarrow$ Centre $S' \equiv 0$ is $(0,2)$ and radius $= 1$

$\Rightarrow$ Radius of $S \equiv 0$ is $2 = \sqrt{\dfrac{\lambda^2}{4} + 4 - 3} \Rightarrow \lambda = \pm 2$

$\Rightarrow$ Distance between their centres $= 1$ unit

55. Equation of tangent having slope m for $y^2 = 8x$ is $y = mx + \dfrac{2}{m}$

$\Rightarrow m^2x - my + 2 = 0$ .....(1)

Equation of tangent to the ellipse with parametric angle θ is

$\dfrac{x\cos\theta}{2\sqrt{k}} + \dfrac{y\sin\theta}{\sqrt{k}} = 1$ .....(2)

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For common tangent (1) and (2) are identical

$\Rightarrow m^2 = \frac{-\cos\theta}{\sqrt{k}}$ and $m = \frac{2\sin\theta}{\sqrt{k}} \Rightarrow \frac{-\cos\theta}{\sqrt{k}} = \frac{4\sin^2\theta}{k}$

$\Rightarrow \sqrt{k} = -\frac{4\sin^2\theta}{\cos\theta} = 6$, for $\theta = \frac{2\pi}{3}$

$\Rightarrow k = 36$

56. Let $y = 2x^2 + 6x + 5$

$y' = 4x + 6$

$4a + 6 = -2$

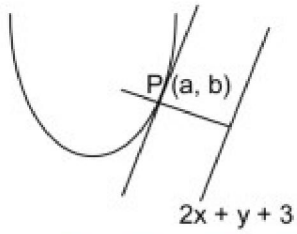
$4a = -8$

$a = -2$

$y = 2(-2)^2 + 6(-2) + 5 = 8 - 12 + 5 = 1$

$P(-2,1)$ equation of normal at P

$y - 1 = \frac{1}{2}(x + 2); 2y - 2 = x + 2; x - 2y + 4 = 0$



57. ${}^7P_4 \times 1 + \frac{{}^7P_3}{2} + 1 - 2 = 943$

58. $P = \frac{\frac{1}{4} \times {}^3c_1 \times \frac{1}{8}}{{}_{c_3} \times \frac{1}{32}} = \frac{3}{10}.$

59. $f(x) = 5 - (5 - x)^2 \Rightarrow (5 - x)^2 = 5 - f(x)$

$f(f(x)) = 5\{5 - f(x)\}^2 = 5 - \{5 - x\}^4$

$f(f(f(x))) = 5 - \{5 - (f(x))\}^4 = 5 - \{5 - x\}^8$

$\int_0^5 \left[5 - \{5 - x\}^8\right] dx = \left[5x - \frac{(5 - x)^{8+1}}{9(-1)}\right]_0^5$

$= 25 - \left\{\frac{5^8}{9}\right\} = \frac{45 - 5^8}{9}$

$= 5^2\left(1 - \frac{5^7}{9}\right)$

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60. S1: $P(B/A)=1$

$\Rightarrow \frac{P(A\cap B)}{P(A)}=1\Rightarrow P(A\cap B)=P(A)$

$\Rightarrow ACB$

S2: $\Rightarrow \frac{P(A\cap B)}{P(B)}\Rightarrow P(A)\Rightarrow \frac{P(A\cap B)}{P(A)}\Rightarrow P(B)[T]$

S3: $P(B)=P(A\cap B)\Rightarrow BCA$

$P(A/B)=\frac{P(A\cap B)}{P(B)}=1$

61. $[x]<x\Rightarrow 2x^2-x[x]\geq 2x^2-x^2=x^2$

$5\geq x^2\Rightarrow x\in\left[-\sqrt{5},\sqrt{5}\right].$

Case (1) $[x]=0\Rightarrow 2x^2=5\Rightarrow x^2=\frac{5}{2}\Rightarrow x=\sqrt{\frac{5}{2}}$

$0\leq x<1$ [Reject]

Case(2) $[x]=1\Rightarrow 1\leq x<2$

$2x^2-x-5=0\Rightarrow x=\frac{1+\sqrt{41}}{4}$

Case: 3: $[x]=2;2\leq x<3$

$2x^2-2x-5=0\ \Rightarrow x=\frac{1+\sqrt{11}}{2}$

Sum of sol: $\frac{1+\sqrt{11}}{2}+\frac{1+\sqrt{41}}{4}=\frac{3+2\sqrt{11}+\sqrt{41}}{4}$

62. $I=\int\frac{d}{dx}\left[\sec^3 2x\cdot\sin^4 3x\cdot e^{5x}\right]dx$

$=\sec^3 2x\cdot\sin^4 3x\cdot e^{5x}+c$

63. S2 is false because points can be collinear also

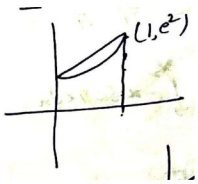
64. $\int_{-1}^{3/2}\left|x\sin(\pi x)\right|dx=\int_{-1}^{1}x\sin\pi x dx+\int_{1}^{3/2}-x\cdot\sin\pi x dx\quad=\frac{3}{\pi}+\frac{1}{\pi^2}\cdot$

$\int_{-1}^{3/2}\left[x^3\right]dx=\int_{-1}^{0}-dx+\int_{0}^{\sqrt[3]{3}}0+\int_{1}^{\sqrt[3]{2}}dx+\int_{\sqrt[3]{2}}^{3/2}2dx+\int_{\sqrt[3]{3}}^{3}3\cdot dx$

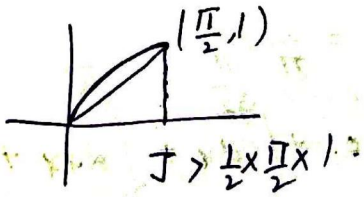
$=-1+0+\sqrt[3]{2}-1+\sqrt[3]{3}-\sqrt[3]{2}+\frac{3}{2}-\sqrt[3]{3}=-1$

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65.



$\therefore I \leq \frac{1}{2}\left(1+e^2\right) < \frac{1}{2}(1+8) = 4\cdot 5$



66. $(x+a)^2+(y-a)^2 = a^2 - 0 : a > 0$

$x + y\frac{dy}{dx} = a\left(\frac{dy}{dx} - 1\right) \Rightarrow a = \frac{x+yy'}{y'-1}$ ___(2)

from (1) & (2)

$(x+y)^2\left[(y')^2+1\right] = \left[x+yy'\right]^2.$

67. Solution of $D\cdot E$.

$y\cdot\sec^2 x = \sec x + c.$

$y\left(\frac{\pi}{3}\right) = 0 \Rightarrow \cot 2\ c = -2$

$y = \cos x - \frac{1}{2}\cos^2 x \quad y = \cos x - 2\cos^2 x$

$y = \frac{1}{8} - \left(\cos x - \frac{1}{4}\right)^2$

$A = \left|\int_0^{\frac{\pi}{3}} \left(\cos x - 2\cos^2 x\right) dx\right| = \left|\int_0^{\pi/3} \left[\cos x - (1+\cos 2x)\right] d.\right.$

$A = \ A = (\sin x)_0^{\pi/3} - \left\{x + \frac{\sin 2x}{2}\right\}_0^{\pi/3}$

$A = \left|\frac{\sqrt{3}}{2} - \left\{\frac{\pi}{3} + \frac{\sqrt{3}}{4}\right\}\right| = \left|-\frac{\pi}{3} + \frac{\sqrt{3}}{4}\right|$

71. $\alpha + \frac{1}{\alpha} \geqslant 2$ a $\alpha - \beta - \frac{1}{\beta} \leqslant 0 \Rightarrow$ one root $\geqslant 2$ ather $\leqslant 0$



$f(0) \leqslant 0 df(2) \leqslant 0 \Rightarrow a \in [-1,3]$

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72. Clearly, variable chord passes through the fixed point $(2,2)$ lying on the circle

$\angle APB = \frac{\pi}{4}$

$\Rightarrow \angle AOB = \frac{\pi}{2}$

$\Rightarrow$ Chord subtends right angle at the origin

Now on homogenization, we get $x^2 + y^2 - 8\left(\frac{mx-y}{2m-2}\right)^2 = 0$

$\Rightarrow (m-1)^2\left[x^2+y^2\right] - 2\left[m^2x^2 - 2mxy + y^2\right] = 0$

$\Rightarrow$ Coefficient of $x^2 +$ coefficient of $y^2 = 0$

$\Rightarrow (m-1)^2 - 2m^2 + (m-1)^2 - 2 = 0$

$\Rightarrow om^2 - 4m + 0 = 0 \Rightarrow m = 0$

73. Given equation of curve $(x-1)(y-2) = 4$; hyperbola

Let $x-1 = X$ and $y-2 = Y \Rightarrow XY = 4$

Clearly normal at $\left(ct, \frac{c}{t}\right)$ intersects the curve again at $\left(ct', \frac{c}{t'}\right)$ where $t' = -\frac{1}{t^3}; c = 2$

$\Rightarrow 2t = 2 \Rightarrow t = 1 \Rightarrow Q = (-2,-2) \Rightarrow Q$ in old system $(\alpha+1, \beta+1) \equiv (-1,0)$

$\Rightarrow \frac{\alpha}{\beta} = 2$

74. $x_1x_2x_3x_4x_5 = 9\times8\times7\times5\times5$

$\Rightarrow$ total numbers $= \frac{5!}{2} = 60$ are same.

75. Let $x-d, x, x+d$ one the sequence

So, it is given that $(x-d)^2 + x^2 + (x+d)^2 = xd^2 \Rightarrow 3x^2 + 2d^2 - xd^2 = 0 \Rightarrow 3x^2 - xd^2 + 2d^2 = 0$

$\Rightarrow x = \frac{d^2 \pm \sqrt{d^4 - 24d^2}}{6} = \frac{d^2 \pm d\sqrt{d^2-24}}{6}$

So, $d = 5$ and $d = 7$ are the only ($+ve$) integers value of d in possible, so that $\sqrt{d^2-24}$ a positive integer.

At $d = 5, x = 5$ and $x = \frac{10}{3}, d = 7, x = 14$ so, $(5+5)+(7+14) = 31$

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